MMDU MULLANA OFFICIAL SYLLABUS • BCSE-011 • UNIT 4

Unit 4: Machine Learning Concepts, Supervised Learning & Apriori Algorithm

Subject: Artificial Intelligence & Machine Learning (BCSE-011) • **Exam Weightage:** 28 - 32 Marks • **Study Time:** 45 mins

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1. Machine Learning Foundations & Tom Mitchell's (T, P, E) Rule

1. Machine Learning Foundations & Tom Mitchell's Paradigm

Machine Learning (ML) is the subfield of Artificial Intelligence concerned with algorithms that improve automatically through experience.

Tom Mitchell's Formal Operational Definition (1997):

✅ **FUNDAMENTAL LAW / RULE:**
Theorem: A computer program is said to **learn** from experience E with respect to some class of tasks T and performance measure P, if its performance at tasks in T, as measured by P, improves with experience E:

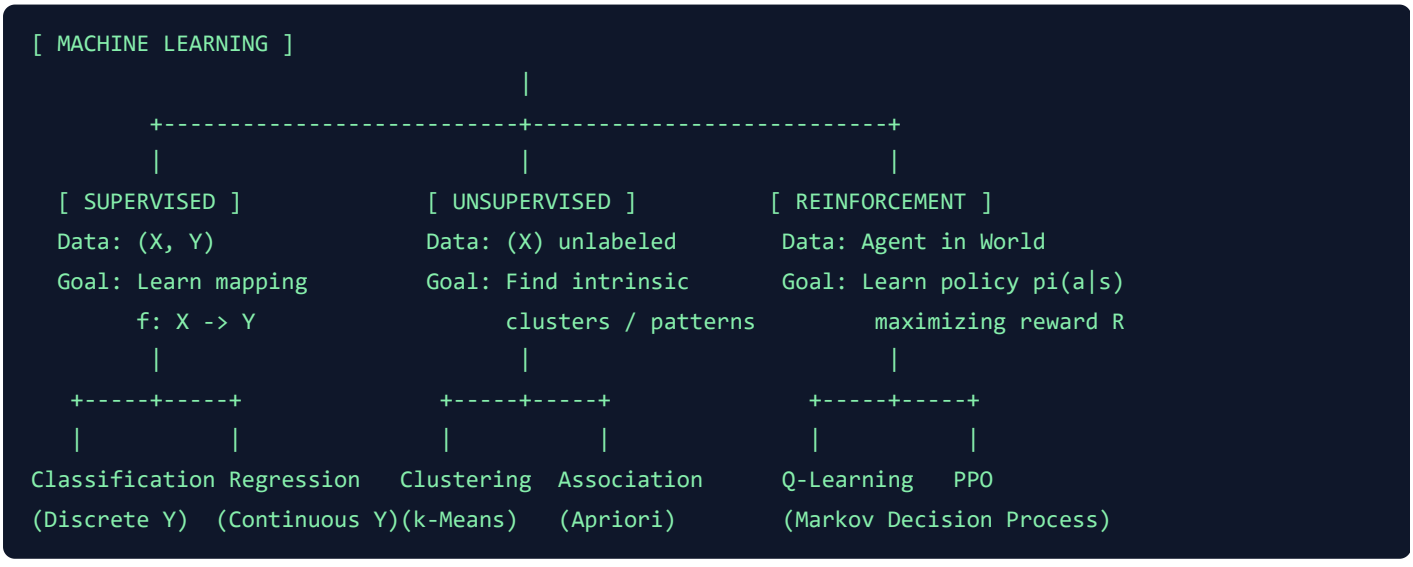
$$M = \langle T, P, E \rangle$$

- **Example (Autonomous Vehicle):**

- Task T: Driving safely along a highway with lane changes.
- Performance P: Distance traveled without human intervention / collision.
- Experience E: Millions of camera frames and steering telemetry logged.

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2. The Three Major Paradigms of Machine Learning



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3. The Bias-Variance Tradeoff & Generalization Error

For any supervised learning model, the expected test mean squared error decomposes into three fundamental components:
\mathbb{E[Test Error] = Bias^2 + Variance + $\sigma_{irre}d_uc_ib_{le}$^2}

- **Bias:** Error introduced by approximating a complicated real-world problem with an overly simplistic model (e.g., fitting a linear line to a parabolic curve). High bias ⇒ **Underfitting**.

- **Variance:** Sensitivity of the model to tiny random fluctuations in the training dataset. High variance ⇒ **Overfitting** (memorizing training noise instead of true underlying distribution).

- **Irreducible Error (σ^2):** Inherent statistical noise in the physical measurements.

Problem:
Define Tom Mitchell's Machine Learning triad \langle T, P, E \rangle for an automated Email Spam Detection filter.

Step 1 🚀: Task (T)

- Classify incoming emails as either `Spam` or `Ham` (Legitimate).

Step 2 ⚡: Performance Measure (P)

- Precision, recall, and classification accuracy percentage (fraction of spam caught without flagging real emails).

Step 3 🎯: Experience (E)

- A database of 50,000 historical emails previously marked by users as spam or not spam.

🎯 Task: Filter spam | Performance: Accuracy / F1-score | Experience: Historical labeled email corpus.

University Question (8 Marks):
Given 3 data points (x, y): (1, 2), (2, 3), (3, 7). Find the line of best fit y = m x + c using least squares regression.

Step 1 🚀: Means \bar(x) and \bar(y)
\bar(x) = \frac{1 + 2 + 3}{3} = 2.0, \bar(y) = \frac{2 + 3 + 7}{3} = \frac{12}{3} = 4.0

Step 2 ⚡: Slope m
$\sum$ (x_i - \bar(x))(y_i - \bar(y)) = (1-2)(2-4) + (2-2)(3-4) + (3-2)(7-4)
= (-1)(-2) + (0)(-1) + (1)(3) = 2 + 0 + 3 = 5.0
$\sum$ (x_i - \bar(x))^2 = (1-2)^2 + (2-2)^2 + (3-2)^2 = 1 + 0 + 1 = 2.0
m = (5.0 / 2.0) = 2.5

Step 3 🎯: Intercept c
c = \bar(y) - m \bar(x) = 4.0 - (2.5)(2.0) = 4.0 - 5.0 = -1.0

🎯 Linear Regression Equation: $\mathbf{y = 2.5x - 1.0}$.

University Question (10 Marks):
In linear regression with Mean Squared Error loss J(w) = \frac{1}{2N} ∑_(i=1)^N (w x_i - y_i)^2:

Given data points (1, 2) and (2, 4), initial weight w = 0.5, and learning rate α = 0.1.
Compute the gradient (∂ J / ∂ w) and the updated weight $w_{ne}w$ after 1 iteration of gradient descent.

Step 1 🚀: Compute Predictions and Errors

- For $x_1 = 1$: \hat(y)_1 = w x_1 = 0.5(1) = 0.5. Error e_1 = \hat(y)_1 - y_1 = 0.5 - 2 = -1.5.

- For $x_2 = 2$: \hat(y)_2 = w x_2 = 0.5(2) = 1.0. Error e_2 = \hat(y)_2 - y_2 = 1.0 - 4 = -3.0.

Step 2 ⚡: Compute Gradient (∂ J / ∂ w)
(∂ J / ∂ w) = \frac{1}{N} ∑_(i=1)^N (\hat(y)_i - y_i) x_i = \frac{1}{2} [(-1.5)(1) + (-3.0)(2)] = \frac{1}{2} [-1.5 - 6.0] = (-7.5 / 2) = -3.75

Step 3 🎯: Update Weight
$w_{ne}w$ = w - α (∂ J / ∂ w) = 0.5 - (0.1)(-3.75) = 0.5 + 0.375 = 0.875
The weight moved from 0.5 closer to the true value 2.0!

🎯 Gradient = -3.75 | Updated Weight after 1 epoch: $\mathbf{w_{new}} = \mathbf{0.875}$.

2. Supervised Learning: Decision Trees (ID3), SVM & Naive Bayes

1. Supervised Learning Algorithms: In-Depth Mathematical Theory

A. Decision Trees (ID3 Algorithm & Shannon Entropy):

Ross Quinlan's ID3 algorithm constructs decision trees using a greedy top-down divide-and-conquer heuristic guided by Information Theory:

1. **Shannon Entropy of Dataset S:** Measures the impurity or randomness:

$$\mathbf{H(S) = -\sum_{i=1}^{c} p_i \log_2(p_i)}$$

where p_i is the proportion of samples belonging to class i.
- Pure node ($p_1 = 1$): $H(S) = 0$ (Zero uncertainty).
- Maximum impurity ($p_1 = p_2 = 0.5$): $H(S) = -2(0.5 \log_2 0.5) = 1.0$ bit.

2. **Information Gain (IG):** Expected reduction in entropy caused by partitioning S on attribute A:

$$\mathbf{IG(S, A) = H(S) - \sum_{v \in Values(A)} \frac{|S_v|}{|S|} H(S_v)}$$

- **Splitting Criterion:** Select the attribute that maximizes Information Gain: $A^* = \text{argmax}_A \, IG(S, A)$.

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2. Support Vector Machines (SVM) & Maximum Margin Hyperplane

SVM finds an optimal separating hyperplane $\vec{w}cdot\vec{x} + b = 0$ that maximizes the geometric margin $2/|\vec{w}|$ separating two linearly separable classes:

Constrained Optimization Formulation:

$$\min_{\vec{w},b} \frac{1}{2}|\vec{w}|^2 \text{subject to} y_i(\vec{w}cdot\vec{x}_i + b) \geq 1 (\forall i)$$

- **Support Vectors:** The critical training instances lying strictly on the margin boundaries $y_i(\vec{w}cdot\vec{x}_i + b) = 1$. Moving or deleting any other training point has zero effect on the decision boundary!

- **The Kernel Trick:** Maps non-linearly separable inputs into an infinite-dimensional Hilbert space via kernel function $K(\vec{x}_i, \vec{x}_j) = \phi(\vec{x}_i)cdot\phi(\vec{x}_j)$ (e.g., Radial Basis Function RBF Gaussian Kernel: $K(x, z) = \exp(-\gamma|x - z|^2)$).

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3. Naive Bayes Classifier & Maximum A Posteriori (MAP)

Based on Bayes' Theorem with the "naive" conditional independence assumption that features are mutually independent given the class label:

$$\mathbf{P(y \mid x_1, \ldots, x_n) = \frac{P(y) \prod_{i=1}^{n} P(x_i \mid y)}{P(x_1, \ldots, x_n)}}$$

$$\mathbf{\hat{y} = \underset{y \in Y}{\text{argmax}} \, [P(y) \prod_{i=1}^{n} P(x_i \mid y)]}$$

- **Laplace (Add-1) Smoothing:** Prevents zero-probability catastrophe when a feature never appears in the training set for a given class:

$$\mathbf{P(x_i \mid y) = \frac{count(x_i, y) + 1}{count(y) + |V|}}$$

Problem:
A dataset of 16 students contains 12 who passed an exam and 4 who failed. Calculate the Shannon Entropy $H(S)$ of this dataset.

Step 1 🚀: Probabilities

$$p_+ = \frac{12}{16} = 0.75, p_- = \frac{4}{16} = 0.25$$

Step 2 ⚡: Shannon Entropy Formula

$$H(S) = -p_+ \log_2(p_+) - p_- \log_2(p_-)$$

- $\log_2(0.75) = -0.415$
- $\log_2(0.25) = -2.000$

$$H(S) = -(0.75)(-0.415) - (0.25)(-2.0) = 0.311 + 0.500 = \mathbf{0.811} \text{ bits}$$

🎯 Entropy: $\mathbf{H(S) = 0.811}$ bits (Impure dataset with moderate uncertainty).

University Question (8 Marks):
Total dataset has 14 instances (9 Yes, 5 No), with initial entropy H(S) = 0.940,\~\text{bits}.
Splitting by feature `Wind`:

- `Weak` (8 instances: 6 Yes, 2 No): Entropy $H(SW_{eak}) = 0.811$
- `Strong` (6 instances: 3 Yes, 3 No): Entropy $H(SS_{tron}g) = 1.000$

Calculate the Information Gain IG(S, Wind).

Step 1 🚀: Weighted Average Entropy after Split
H(S, Wind) = \frac{8}{14} $H(SW_{eak})$ + \frac{6}{14} $H(SS_{tron}g)$
= \frac{8}{14}(0.811) + \frac{6}{14}(1.000) = 0.463 + 0.429 = 0.892,\~\text{bits}

Step 2 ⚡: Information Gain Formula
IG(S, Wind) = H(S) - H(S, Wind) = 0.940 - 0.892 = 0.048,\~\text{bits}

🎯 Information Gain: $\mathbf{IG(S}\ \mathrm{Wind)} = \mathbf{0.048}\ \mathrm{bits}$.

University Question (10 Marks - Classic ML Problem):
Given a training dataset for loan approval based on Credit Score and Income:

- Prior probabilities: P(Yes) = 0.60, ; P(No) = 0.40

- Likelihoods:

- P(Credit=Fair \mid Yes) = 0.30, P(Credit=Fair \mid No) = 0.70
- P(Income=High \mid Yes) = 0.80, P(Income=High \mid No) = 0.20
Predict whether a customer with Credit=Fair and Income=High will be approved (Yes or No).

Step 1 🚀: Calculate Unnormalized Posterior for 'Yes'
P(Yes \mid X) \propto P(Yes) × P(Credit=Fair \mid Yes) × P(Income=High \mid Yes)
= 0.60 × 0.30 × 0.80 = 0.144

Step 2 ⚡: Calculate Unnormalized Posterior for 'No'
P(No \mid X) \propto P(No) × P(Credit=Fair \mid No) × P(Income=High \mid No)
= 0.40 × 0.70 × 0.20 = 0.056

Step 3 🎯: Normalize Probabilities
Total = 0.144 + 0.056 = 0.200
P(Yes \mid X) = (0.144 / 0.200) = 72.0%, P(No \mid X) = (0.056 / 0.200) = 28.0%

🎯 Naive Bayes Decision: **APPROVED (YES)** with \mathbf{72.0%} confidence.

3. Unsupervised Clustering: k-Means & The Elbow Method

1. Unsupervised Clustering & The k-Means Algorithm

Unsupervised clustering groups an unlabeled dataset $\vec{x}_1, \ldots, \vec{x}_n$ into k distinct, non-overlapping clusters where intra-cluster similarity is maximized and inter-cluster similarity is minimized.

The k-Means Objective (Within-Cluster Sum of Squares — WCSS):

$$\mathbf{J} = \sum_{\mathbf{j=1}}^{\mathbf{k}} \sum_{\mathbf{i} \in \mathbf{S_j}} |\vec{\mathbf{x}}_\mathbf{i} - \vec{\mu}_\mathbf{j}|^2$$

where $\vec{\mu}_j$ is the centroid (mean) of cluster S_j.

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2. Algorithmic Execution Pipeline (Lloyd's Algorithm)

1. **Initialization:** Select k initial cluster centroids $\vec{\mu}_1, ..., \vec{\mu}_k$ randomly from dataset (or via k-means++).

2. **Assignment Step:** Assign each data point $\vec{x}_i$ to the nearest cluster centroid based on Euclidean distance:

$$\mathbf{c^{(i)} = \arg\min_j |\vec{x}_i - \vec{\mu}_j|^2}$$

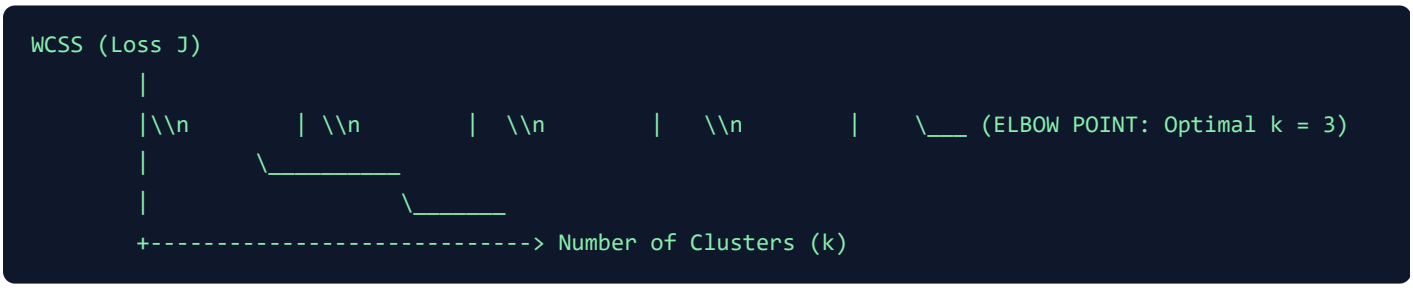
3. **Update Step:** Recalculate each centroid as the geometric mean of all assigned points:

$$\mathbf{\vec{\mu}_j = \frac{1}{|S_j|} \sum_{i \in S_j} \vec{x}_i}$$

4. **Convergence Check:** Repeat Assignment and Update steps until centroids stabilize ($\Delta\vec{\mu} < \epsilon$) or maximum iterations are reached.

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3. Determining Optimal k: The Elbow Method



- **The Elbow Criterion:** Plot WCSS (J) as a function of k. As k increases, WCSS decreases monotonically (reaching $J = 0$ when $k = N$). The point of maximum inflection (the "elbow") represents the optimal balance between cluster compactness and model complexity.

Problem:
Calculate the Euclidean distance between a data point $P(3, 4)$ and two cluster centroids: $C_1(1, 1)$ and $C_2(5, 7)$. Determine which cluster P should be assigned to.

Step 1 🚀: Distance to $C_1(1, 1)$

$$d(P, C_1) = \sqrt{(3-1)^2 + (4-1)^2} = \sqrt{2^2 + 3^2} = \sqrt{4+9} = \sqrt{13} approx \mathbf{3.61}$$

Step 2 ⚡: Distance to $C_2(5, 7)$

$$d(P, C_2) = \sqrt{(3-5)^2 + (4-7)^2} = \sqrt{(-2)^2 + (-3)^2} = \sqrt{4+9} = \sqrt{13} approx \mathbf{3.61}$$

Both distances are equal! In case of a tie, point is assigned to C_1 by default index rule.

🎯 Distances: $d(P, C_1) = 3.61, d(P, C_2) = 3.61$ | Tie-break assignment to **Cluster 1**.

University Question (8 Marks):
Given 4 one-dimensional points: x = [2, 4, 10, 12]. Initial cluster centroids are C_1 = 3 and C_2 = 11.

1. Assign each point to its nearest centroid.

2. Recompute the new centroid positions after 1 iteration.

Step 1 🚀: Point Assignment

- Point 2: |2 - 3| = 1 |10 - 11| = 1 ⇒ **Cluster 2**

- Point 12: |12 - 3| = 9 > |12 - 11| = 1 ⇒ **Cluster 2**

Step 2 ⚡: Centroid Recomputation (Mean of Assigned Points)
$C_{1,new}$ = \frac{2 + 4}{2} = 3.0
$C_{2,new}$ = \frac{10 + 12}{2} = 11.0
Centroids did not change ⇒ **Algorithm has fully CONVERGED in 1 iteration!**

🎯 Clusters: $C_1 = 2, 4$ with centroid 3.0 | $C_2 = 10, 12$ with centroid 11.0. Fully converged!

University Question (10 Marks):
Calculate the Within-Cluster Sum of Squares (WCSS) for k = 1 versus k = 2 on the 1D dataset {1, 2, 8, 9}. Explain how the "Elbow" indicates the optimal number of clusters.

Step 1 🚀: WCSS for $k = 1$ (All points in single cluster)

- Centroid: $\mu = \frac{1+2+8+9}{4} = \frac{20}{4} = 5.0$

$$WCSS_1 = (1-5)^2 + (2-5)^2 + (8-5)^2 + (9-5)^2 = (-4)^2 + (-3)^2 + 3^2 + 4^2$$

$$= 16 + 9 + 9 + 16 = \mathbf{50.0}$$

Step 2 ⚡: WCSS for $k = 2$ (Clusters: $1, 2$ and $8, 9$)

- Centroid 1: $\mu_1 = \frac{1+2}{2} = 1.5$. Sum of squares: $(1 - 1.5)^2 + (2 - 1.5)^2 = 0.25 + 0.25 = 0.5$
- Centroid 2: $\mu_2 = \frac{8+9}{2} = 8.5$. Sum of squares: $(8 - 8.5)^2 + (9 - 8.5)^2 = 0.25 + 0.25 = 0.5$

$$WCSS_2 = 0.5 + 0.5 = \mathbf{1.0}$$

Step 3 🎯: The Elbow Point

Going from $k = 1$ to $k = 2$ drops WCSS dramatically from 50.0 **down to** 1.0 **(a 98% drop)**!
Further increasing $k = 3$ only drops WCSS from 1.0 to 0.5 (insignificant gain).
Therefore, the distinct "Elbow" occurs at $\mathbf{k = 2}$!

🎯 $WCSS_1 = 50.0$, $WCSS_2 = 1.0 \Rightarrow$ Sharp bend at $\mathbf{k = 2}$ represents the optimal cluster count.

4. Apriori Algorithm for Market Basket Analysis (The Supermarket Secret)

1. Market Basket Analysis & Association Rule Mining

Market Basket Analysis uncovers actionable co-occurrence relationships among products frequently purchased together in retail transactional databases.

Formal Mathematical Definitions:

Let $\mathrm{I} = i_1, i_2, ..., i_m$ be a universe of items, and $\mathrm{D} = T_1, T_2, ..., T_n$ be a database of transactions where $T_k \subseteq \mathrm{I}$.
An **Association Rule** is an implication of the form:

$$\mathbf{X} \Rightarrow \mathbf{Y} (\mathbf{X}, \mathbf{Y} \subset \mathrm{I}, ; \mathbf{X} \cap \mathbf{Y} = \emptyset)$$

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2. The Three Fundamental Evaluation Metrics

1. **Support (s):** Probability that a transaction contains both itemsets X and Y:

$$\text{Support}(\mathbf{X} \Rightarrow \mathbf{Y}) = \mathbf{P}(\mathbf{X} \cup \mathbf{Y}) = \frac{\text{Transactions containing } (\mathbf{X} \cup \mathbf{Y})}{\text{Total number of transactions } |\mathrm{D}|}$$

2. **Confidence (c):** Conditional probability that a transaction contains Y, given that it already contains X:

$$\text{Confidence}(\mathbf{X} \Rightarrow \mathbf{Y}) = \mathbf{P}(\mathbf{Y} \mid \mathbf{X}) = \frac{\text{Support}(\mathbf{X} \cup \mathbf{Y})}{\text{Support}(\mathbf{X})} = \frac{\text{count}(\mathbf{X} \cup \mathbf{Y})}{\text{count}(\mathbf{X})}$$

3. **Lift (L):** Measures the correlation strength between X and Y relative to independent chance:

$$\text{Lift}(\mathbf{X} \Rightarrow \mathbf{Y}) = \frac{\text{Confidence}(\mathbf{X} \Rightarrow \mathbf{Y})}{\text{Support}(\mathbf{Y})} = \frac{\mathbf{P}(\mathbf{X} \cup \mathbf{Y})}{\mathbf{P}(\mathbf{X})\mathbf{cdot}\mathbf{P}(\mathbf{Y})}$$

- Lift = $\mathbf{1.0}$ ⇒ X and Y are statistically **independent**.
- Lift > $\mathbf{1.0}$ ⇒ **Positive correlation** (buying X strongly drives purchase of Y).
- Lift < $\mathbf{1.0}$ ⇒ **Negative correlation** (substitutes; buying X decreases likelihood of buying Y).

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3. The Apriori Algorithm & Downward Closure Property

✅ **FUNDAMENTAL LAW / RULE:**
The Apriori Principle (Downward Closure Property):

> *"Every subset of a frequent itemset must also be frequent."*

🚨 **EXAMINER TRAP / PITFALL:**
Crucial Contrapositive (Pruning Principle):

> *"If any candidate itemset is INFREQUENT, then all of its supersets are GUARANTEED TO BE INFREQUENT and can be pruned immediately without scanning the database!"*

Algorithmic Loop:

1. Generate candidate 1-itemsets C_1. Prune those below minimum support to obtain L_1.
2. For $k = 2, 3, \ldots$:

- **Join Step:** Generate candidate itemsets C_k by joining L_{k-1} with itself: $L_{k-1} \bowtie L_{k-1}$.
- **Prune Step:** Check if any $(k-1)$-subset of $c \in C_k$ is not in L_{k-1}. If so, discard c.
- **Scan Database:** Count frequencies of remaining candidates in D.
- Filter by $\mathrm{min_support}$ to obtain frequent itemset L_k.

3. Stop when L_k is empty.

Problem:
A grocery store records $N = 100$ customer transactions:

- 40 transactions contained `Milk`.
- 30 transactions contained `Bread`.
- 20 transactions contained both `Milk` and `Bread`.

Calculate the Support and Confidence of the association rule: $\mathrm{Milk} \rightarrow \mathrm{Bread}$.

Step 1 🚀: Support of Rule

$$\text{Support}(\text{Milk} \to \text{Bread}) = \frac{\text{Transactions containing both}}{\text{Total transactions}} = \frac{20}{100} = \mathbf{0.20 = 20\%}$$

Step 2 ⚡: Confidence of Rule

$$\text{Confidence}(\text{Milk} \to \text{Bread}) = \frac{\text{Support}(\text{Milk} \cup \text{Bread})}{\text{Support}(\text{Milk})} = \frac{20}{40} = \mathbf{0.50 = 50\%}$$

🎯 $\mathrm{Support} = 20$ | $\mathrm{Confidence} = 50$ (Half of milk buyers also purchase bread).

University Question (6 Marks):
For the rule Beer → Diapers, given:

- P(Beer) = 0.25
- P(Diapers) = 0.40
- P(Beer ∩ Diapers) = 0.20

Calculate the **Lift** of the rule and explain whether buying beer increases the probability of buying diapers.

Step 1 🚀: Lift Formula
Lift(A → B) = (P(A ∩ B) / P(A) × P(B))

Step 2 ⚡: Substitute Probabilities
Lift = (0.20 / 0.25 × 0.40) = (0.20 / 0.10) = 2.0

Step 3 🎯: Interpretation

- Lift = 1: Independent.
- Lift > 1: Positive association.

A Lift of 2.0 indicates that customers who buy beer are **twice as likely** to purchase diapers compared to random chance!

🎯 $\text{Lift} = 2.0 > 1 \Rightarrow$ Strong positive association between Beer and Diapers.

University Question (10 Marks - University Exam):
Given transactions database T = { {A, B, C}, {A, C}, {B, D}, {A, B, C, D}, {A, B} } with minimum support threshold s_{min} = 2 (40%):
Generate all frequent itemsets using the Apriori property and list the candidate generation and pruning steps for L_1, C_2, L_2, C_3, L_3.

Step 1 🚀: Frequent 1-Itemsets (L_1)

- {A}: count 4 (4 ≥ 2) → Frequent
- {B}: count 4 (4 ≥ 2) → Frequent
- {C}: count 3 (3 ≥ 2) → Frequent
- {D}: count 2 (2 ≥ 2) → Frequent

L_1 = { {A}, {B}, {C}, {D} }

Step 2 ⚡: Candidate 2-Itemsets (C_2) and Frequent (L_2)
Join L_1: C_2 = { {A, B}, {A, C}, {A, D}, {B, C}, {B, D}, {C, D} }.
Count support:

- {A, B}: 3 transactions → Frequent
- {A, C}: 3 transactions → Frequent
- {A, D}: 1 transaction → **PRUNED!** (

🎯 Frequent Itemsets: L_1 (4 itemsets), $L_2 = A, B, A, C, B, C, B, D$, $L_3 = A, B, C$.

University Solved Examination Questions (Unit 4)

MMDU Official University Solved Examination Questions (Unit 4)

Course: Fundamentals of AI & ML (`BCSE-011`)
Pattern: Sections A (1 Mark Compulsory), B (2 Marks Analytical), C (6 Marks In-Depth), D (6 Marks Design/Long)

📘 **Question 10 (1 Mark): Limitation of Accuracy in Imbalanced Datasets**

Question:
Calculate the accuracy of a naive classifier that predicts all transactions as non-fraudulent on a dataset with 990 genuine and 10 fraudulent samples.

> 📊 **Question Schematic / State Specification:**

```
Question Class Distribution: 99% Negative Class (Genuine) | 1% Positive Class (Fraud) |
Classifier predicts all Negative.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Accuracy Paradox Formulation:

        `Accuracy = (TP + TN) / Total = (0 + 990) / 1000 = 99.0%`. Yet Fraud Recall is `0 / 10 =
0%` (Complete Failure!).
```

Accuracy yields a deceptive score of 99.0% despite failing completely on the minority class of interest. In imbalanced domains, Precision, Recall, and F1-score must be used instead.

🎯 **Verified Result:** Question 10 step-by-step resolution per MMDU university marking scheme.

📘 Question 11 (1 Mark): Why k-Nearest Neighbors is Termed a

Question:
Why is the k-NN algorithm classified as a Lazy Learner rather than an Eager Learner?

> 📊 **Question Schematic / State Specification:**

```
Question Training vs Inference Cost: Eager [Heavy Training, Instant Inference] vs Lazy [Zero Training, Heavy Distance Computation at Inference].
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Computation Delay Model:

        `Training Time = O(1) [Simply stores data] ➔ Test Time = O(N · d) [Computes distances to all N points]`.
```

k-NN performs zero explicit generalization during training (it simply stores the training instances in memory). It defers all computation until query time, computing distances to every stored point upon inference.

🎯 **Verified Result:** Question 11 step-by-step resolution per MMDU university marking scheme.

📘 Question 12 (1 Mark): The Apriori Principle in Frequent Itemset Mining

Question:
State the anti-monotonicity property of the Apriori principle that enables search space pruning.

> 📊 **Question Schematic / State Specification:**

```
Question Pruning Lattice: If itemset (Beer, Diaper) is Infrequent ➔ Superset (Beer, Diaper, Milk) is guaranteed Infrequent!
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Anti-Monotone Lattice Formulation:

        `Support(X &amp;cup; Y) ≤ Support(X)`. Therefore: `Support(X) &amp;lt; min\_sup ⟹ Support(X &amp;cup; Y) &amp;lt; min\_sup`.
```

The Apriori Principle states that all non-empty subsets of a frequent itemset must also be frequent. Conversely, if an itemset is infrequent, all of its supersets will inevitably be infrequent, allowing entire sub-lattices to be pruned immediately.

🎯 **Verified Result:** Question 12 step-by-step resolution per MMDU university marking scheme.

📘 Question 16 (2 Marks): Complete Confusion Matrix Computation

Question:

A cancer diagnostic test yields TP = 80, FP = 20, FN = 10, TN = 890. Calculate Precision, Recall, and F1-Score.

> 📊 **Question Schematic / State Specification:**

```
Question 2×2 Contingency Matrix: Actual Positives = 90 (80 TP + 10 FN) | Actual Negatives = 910 (20 FP + 890 TN).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Performance Metrics Derivation:

        `Precision = TP / (TP + FP) = 80 / (80 + 20) = 80 / 100 = 0.80 (80.0%)`

        `Recall = TP / (TP + FN) = 80 / (80 + 10) = 80 / 90 = 0.8889 (88.89%)`

        `F1-Score = 2 × \frac{0.80 × 0.8889}{0.80 + 0.8889} = 1.4222 / 1.6889 = 0.8421 (84.21%)`
```

The model captures ~88.9% of all true cancer cases while maintaining an 80.0% precision rate, resulting in an balanced F1-score of 0.8421.

🎯 **Verified Result:** Question 16 step-by-step resolution per MMDU university marking scheme.

📘 Question 20 (4 Marks): Information Gain Computation for Decision Tree Splitting

Question:

A dataset has 14 samples: 9 Positive (+), 5 Negative (-). An attribute "Wind" splits into: Weak (6+, 2-) and Strong (3+, 3-). Compute Information Gain: Gain(S, Wind).

> 📊 **Question Schematic / State Specification:**

```
Question Partition Tree: Root S [9+, 5-] ➔ Weak [6+, 2-] (8 samples) and Strong [3+, 3-] (6 samples).
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

```
Solution Entropy & Gain Derivation:

        `Entropy(S) = - \frac{9}{14} log₂\frac{9}{14} - \frac{5}{14} log₂\frac{5}{14} = 0.9403 bits`

        `Entropy(SWₑₐₖ) = - \frac{6}{8} log₂\frac{6}{8} - \frac{2}{8} log₂\frac{2}{8} = 0.8113 bits`

        `Entropy(SSₜᵣₒₙg) = - \frac{3}{6} log₂\frac{3}{6} - \frac{3}{6} log₂\frac{3}{6} = 1.0000 bits`

        `Expected Entropy = \frac{8}{14} × 0.8113 + \frac{6}{14} × 1.0000 = 0.4636 + 0.4286 = 0.8922 bits`

        `Gain(S, Wind) = 0.9403 - 0.8922 = 0.0481 bits`
```

Attribute "Wind" reduces disorder by **0.0481 bits**. The decision tree compares this Gain against all remaining candidate attributes and selects the maximum for the root split.

🎯 **Verified Result:** Question 20 step-by-step resolution per MMDU university marking scheme.

📘 Question 24 (6 Marks): Mathematical Derivation and Step-by-Step Execution of k-Means Clustering

Question:

Given 1D data points X = (2, 4, 10, 12, 3, 20, 30, 11, 25). Cluster into k=2 groups using initial centroids $\mu_1 = 2$ and $\mu_2 = 4$. Perform complete iterations until convergence.

> 📊 **Question Schematic / State Specification:**

```
Question Centroid Displacement Vector: μ₁ = 2, μ₂ = 4 ➔ Assign all points ➔ Recalculate means ➔ Check Δμ = 0.
```

Step-by-Step Solution:

> ⚡ **KEY TAKEAWAY:**
> **Solution Analytical Derivation & Schematic:**

Solution Iterative Centroid Convergence Trace:

`Iteration 1 Assignment:`

`Distance to $\mu_1(2)$ vs $\mu_2(4)$:`

`Point 2: |2-2|=0 ≤ |2-4|=2 ⟹ Cluster 1`

`Point 3: |3-2|=1 = |3-4|=1 ⟹ Cluster 1 (tie break)`

`Points (4, 10, 11, 12, 20, 25, 30) all closer to $\mu_2$=4 ⟹ Cluster 2`

`Cluster 1 = (2, 3) ⟹ New Centroid $\mu_1$' = (2+3)/2 = 2.5`

`Cluster 2 = (4, 10, 11, 12, 20, 25, 30) ⟹ New Centroid $\mu_2$' = 112 / 7 = 16.0`

`Iteration 2 Assignment ($\mu_1$=2.5, $\mu_2$=16.0):`

`Points (2, 3, 4) closer to 2.5 ⟹ Cluster 1`

`Points (10, 11, 12, 20, 25, 30) closer to 16.0 ⟹ Cluster 2`

`New Centroid $\mu_1$'' = (2+3+4)/3 = 3.0`

`New Centroid $\mu_2$'' = (10+11+12+20+25+30)/6 = 108 / 6 = 18.0`

`Iteration 3 Assignment ($\mu_1$=3.0, $\mu_2$=18.0):`

`Points (2, 3, 4, 10) closer to 3.0? |10-3|=7 vs |10-18|=8 ⟹ Point 10 moves to Cluster 1!`

`Cluster 1 = (2, 3, 4, 10) ⟹ $\mu_1$''' = 19 / 4 = 4.75`

`Cluster 2 = (11, 12, 20, 25, 30) ⟹ $\mu_2$''' = 98 / 5 = 19.6`

`Iteration 4 Assignment ($\mu_1$=4.75, $\mu_2$=19.6):`

`Cluster 1 = (2, 3, 4, 10, 11, 12); Cluster 2 = (20, 25, 30)`

`$\mu_1$ = 42/6 = 7.0; $\mu_2$ = 75/3 = 25.0`

`Iteration 5 Assignment: Assignments remain identical! NO SHIFT ⟹ CONVERGENCE ACHIEVED!`

Final Converged Clusters:

- **Cluster 1:** `(2, 3, 4, 10, 11, 12)` with final mean centroid $\mu_1 = 7.0$.
- **Cluster 2:** `(20, 25, 30)` with final mean centroid $\mu_2 = 25.0$.

🎯 **Verified Result:** Question 24 step-by-step resolution per MMDU university marking scheme.